



This illustration from a 12th-century manuscript graphically depicts the use of Greek Fire. Flames are being projected from a handheld tube onto a fleet of invading soldiers.

**IN CHINA**, in the year 610 court physician **Chao Yuanfang** (550–630) compiled the first **comprehensive Chinese treatise on diseases**. One of the diseases he described was smallpox; he explained that lesions with purple or black coloration were far more deadly than those that contained white pus. He also recommended brushing teeth daily and proposed a routine of rinsing and gargling then gnashing the teeth seven times.

“AS THE **SUN ECLIPSES THE STARS** BY ITS BRILLIANCY, SO THE **MAN OF KNOWLEDGE** WILL ECLIPSE THE FAME OF OTHERS IN ASSEMBLIES OF THE PEOPLE IF HE PROPOSES **ALGEBRAIC PROBLEMS**, AND STILL MORE IF HE **SOLVES THEM**. ”

**Brahmagupta**, Indian mathematician, from *Brahmasphutasiddhanta* (*The Revised System of Brahma*), 628

Before 644, **windmills had been developed in Persia**. They used wind to drive wooden vanes set in a circle around a windshaft. This generated rotational energy,

which could be used to grind wheat. The earliest windmills had **vertical windshafts**, unlike the more familiar horizontal types that were later developed in Europe.

Spanish bishop **Isidore of Seville** was a prolific author who wrote books on cosmology and arithmetic. In the 7th century, he compiled a **20-volume manuscript of**

**contemporary knowledge**, entitled *Etymologiae*, using the work of earlier encyclopedists such as Roman author Marcus Terentius Varro (116–27 BCE). It helped disseminate classical knowledge in the Middle Ages.

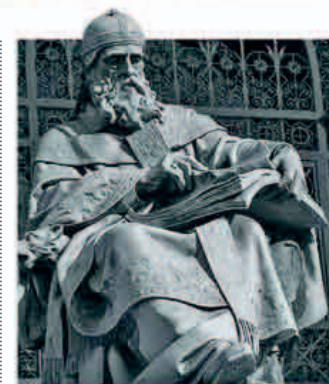
In the field of surgery, Greek physician **Paul of Aegina** (c.625–c.690) compiled *The Epitome of Medicine*—a digest of **medical treatises by ancient authorities** such as Galen. It also contained descriptions of new surgical procedures, such as tracheotomy (surgery to the windpipe) and sterilizing wounds through cauterization.

Chinese mathematician **Wang Xiaotong** (c.580–c.640) was the first to provide **solutions for cubic equations** (of the form  $a^3+ba^2+ca=n$ ). It was a technique that European mathematicians did not master until Fibonacci (see 1220–49) in the 13th century.

In India, one of the greatest early mathematicians was **Brahmagupta** (598–c.668). His *Brahmasphutasiddhanta* (*The Revised System of Brahma*) contained **rules** for using **negative numbers** in arithmetic and also first stated the rule that

#### Vertical windmills

Because the area around Nishtafun in Persia (Iran) experienced high winds, but had little water, windmills were a very useful adaptation.



**ISIDORE OF SEVILLE**  
(c.560–636)

The Bishop of Seville for more than 30 years, Isidore wrote several important texts, including the encyclopedic *Etymologiae*, a dictionary of synonyms, and a manual of basic physics. He also established a system of seminaries to promote ecclesiastical education. He was canonized in 1598 by Pope Clement VIII.

two negative numbers multiplied together yield a positive number.

In the late 7th century, a new incendiary weapon was developed in the Byzantine empire. Known as **Greek Fire**, it was discharged by tubes and burned even in contact with water. Its exact composition is still unknown, but it was probably a compound of naphtha (a hydrocarbon mixture).

c.610 Chao Yuanfang's treatise on disease contains **description of smallpox**

c.625 Wang Xiaotong describes **solution for cubic equations**

c.615–30 Isidore of Seville compiles *Etymologiae*

c.628 First secure use of zero in Indian mathematical texts

644 Vertical windmills first mentioned in Persia (Iran)

c.650 High quality porcelain is invented in China

c.650 Byzantine doctor Theophrastus Protospatharius writes a treatise on urology

c.660 Paul of Aegina describes **breast cancer** and the use of cauterization in surgery

c.670 Greek Fire developed in Byzantine empire